A Literal Factor Dictionary for Prime–Squarefree Orbit Packets: Exact Atomic Normalization and Certified Interior Overlap

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Abstract

The preceding diagonal Mellin reduction isolated an exact-profile dictionary and a positive profile-overlap condition as two application hypotheses. We discharge the first hypothesis, exactly and coordinatewise, for one explicitly declared representative packet from the prime–squarefree opened-row provenance. The packet is formed before coherent prime summation and before the many-to-one grouping $s = d(m)r$. Its orbit multiplier is the literal four-factor product

$$W_1(mj/X)\Psi_1(d(m)j/K)W_2(m_\gamma j/X)\Psi_2(d_\gamma j/K).$$

After direct residue splitting, every factor is assigned to one of four places: the source coefficient, the coefficient-free opposite-row factor, the literal static mask, or the real-axis Mellin measure. The opposite-row logarithmic base weight in the left residual route is exactly 1, not $(\log \ell_\gamma)^2$; its fixed unary shape remains in both profiles and its unit phase remains in the Mellin layer. The single physical prime logarithm remains in the source coefficient. The resulting carrier and physical coordinate identities hold with equality for every X, with all factors of 2π, residue amplitudes, cofactor repetitions, and global phases accounted for. Thus the exact-profile assumption of the preceding paper is a theorem on this declared direct packet.

We next replace the remaining qualitative overlap hypothesis by a deterministic support certificate. In normalized variables $x = \log(\ell_\gamma/L)$, $y = \log(d_\gamma/D)$, and $t = j/J$, the opposite-row factors have arguments

$$c_X e^{x+y} t, \qquad c'_X e^y t, \quad c_X = LDJ/X, \quad c'_X = DJ/K = 1.$$

If at one reachable normalized point the carrier—including the fixed unary shape and retained residue amplitude—is nonzero and all four physical cutoffs lie strictly inside their nonzero sets, then fixed compact source, opposite-row, and orbit boxes can be selected on which the profile-to-carrier ratio has an explicit positive lower bound. The selected opposite-row and orbit boxes have fixed positive geometric density, literal-mask deletion is $o(1)$, and the extraction has zero positive polynomial cost at the $1/400$ endpoint. Once this explicit certificate is verified, the preceding sampled theorem yields an $O(1)$ canonical Mellin condition number for that named retained-label subpacket.

This is not a statement about the whole inherited packet. An arbitrary bounded source factor ζ_m can vanish on every selected compatible source box and concentrate all actual source energy on its complement. Hence positive geometric density does not imply a positive good-carrier fraction. We prove the corresponding sharp obstruction and give a sufficient fixed-nonzero-source criterion for positive source-row energy coverage. Positive coverage after cofactor repetition additionally follows under a balanced-fiber condition, which is not proved here. Coherent prime summation, residual grouping, fixed-shift localization, native-Gram identification, and every parity-sensitive or twin-prime conclusion remain open.

Keywords: Mellin inversion; prime–squarefree rows; orbit fibers; atomic normalization; support compatibility; static mask; sieve parity.

Convention. All residue cells are direct cells, not Fourier characters, unless stated otherwise. The literal static mask is never replaced by a signed rank-one representation. Every norm called "atomic" below is the retained-label pre-terminal norm; it is not the terminal residual norm or the native affine physical Gram. Constants may depend on the fixed cutoffs and residue moduli but not on X.

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1 Introduction

TPC–52 separated a misleading scalar-frame problem from the actual label-preserving Mellin synthesis. On the retained coefficient space it proved that the conditioning of the provenance-fixed direction is governed by a nonnegative physical profile-to-carrier ratio, rather than by a lower frame for every signed Mellin density [7]. Its actual application deliberately retained two hypotheses: a coordinatewise dictionary identifying the abstract profile with the literal opened-row coefficient, and a uniform positive overlap of the physical cutoffs. Neither may be inferred merely from bounded smoothness.

The first gap is bookkeeping, but bookkeeping at this interface is mathematics. There are three different row roles in the provenance chain. The original symmetric opened-row packet has two physical row coefficients. After large-prime descent, a left residual packet has one source coefficient and one coefficient-free opposite output row. After residual grouping, different pairs (m, r) may share the same physical label $s = d(m)r$. Moving a prime logarithm between the first two roles, or declaring (m, r) orthogonal after the third operation, changes the norm being estimated.

This paper fixes one representative four-factor packet and prints its factor allocation in full. No unidentified $\mathcal{W}$ remains in the main theorem. The allocation converts the exact-profile dictionary of TPC–52 from an assumption into an equality on the named packet.

The second gap is geometric. For arbitrary fixed smooth tests, the nonzero sets of W and Ψ need not overlap every reachable scaled orbit. We therefore do not assert that the full parameter box is uniformly active. Instead we prove a necessary-and-sufficient strict support criterion for a nonempty active interior and a quantitative rectangle certificate for a compact subpacket. This certificate is finite-dimensional and deterministic; it contains no prime-pair input.

1.1 Main results

The conclusions are as follows.

(R1) **Exact literal dictionary.** On one complete nondegenerate direct left packet, with no independent opposite-row pruning, we take

$$a_m = \gamma_m^{(1)}, \qquad \omega_{\gamma,X} = 1, \qquad M_X(m, \gamma) = K_{m\gamma}.$$

The fixed opposite unary shape is placed in both the carrier and physical profiles, its unit phase is kept in the layer phase, and the direct residue amplitudes are placed in both profiles. The absolute Mellin transforms and every factor $(2\pi)^{-1}$ are placed in the positive provenance measure. All remaining row and orbit powers have modulus one. The two coordinate identities required in TPC–52 then hold exactly for every retained (a, m, r, j, γ).

(R2) **Atomic normalization.** The exact squared norm is

$$\sum_{m,r,j,\gamma} |\gamma_m^{(1)}|^2 K_{m\gamma} |R(q_\gamma)|^2 |\xi_a|^2 \prod_{\nu=1}^{4} |U_\nu(z_\nu)|^2, \qquad \mathbf{U} = (W_1, \Psi_1, W_2, \Psi_2).$$

It is the norm of a pre-terminal orthogonal lift. The terminal factor $\mu(r)\Lambda(pr)$, coherent prime sum, and collision map $s = d(m)r$ are absent. We prove no equality with the final TPC–48 output norm or native affine Gram.

(R3) **Reachable-support certificate.** A strict point at which the carrier and all four physical factors are nonzero generates compact source, opposite-row, and orbit boxes with explicit overlap

$$\rho_* \geq \frac{r_0^2 \prod_{\nu=1}^{4} w_{\nu,0}^2}{H_0 b_+} \left(\int_{Q} \mathrm{e}^{x+y}\, dx\, dy \right) |I| > 0.$$

Conversely, positive continuous profile energy implies such a strict point. Thus the active parameter set is relatively open and its compact subsets are uniformly active. A grazing limit of active parameters can lose its lower bound at the boundary.

(R4) **A sampled interior bridge.** The row box has limiting geometric mass

$$(\mathrm{e}^{x_+} - \mathrm{e}^{x_-})(\mathrm{e}^{y_+} - \mathrm{e}^{y_-}) > 0,$$

while one orbit residue class contributes $|I|/H_0 + O(J^{-1})$. Fixed-modulus prime and squarefree sampling, together with the literal-mask degree estimate, transfers the explicit continuous overlap to the sampled packet. The canonical actual direction therefore has $O(1)$ condition number on that subpacket.

(R5) **Geometric density is not energy coverage.** If only $|\zeta_m| \ll 1$ is known, the source weight may vanish on every selected active box. We give an exact construction with positive geometric density but zero good-carrier fraction. If instead the remaining source factor is bounded below on a compatible fixed cell, the standard fixed-modulus row asymptotics yield a positive source-energy fraction for that declared finite cell decomposition.

1.2 Claim level

The factor identities and support criterion are Level L0 exact algebra and analysis. Their application to the literal complete row cell and static mask is a Level L1 physical-interface theorem. No coherent prime square, fixed-h_0 arithmetic lower bound, parity-breaking estimate, Hardy–Littlewood asymptotic, or twin-prime conclusion is obtained. In particular, selecting a positive-density interior subpacket is not a substitute for proving that it carries a positive fraction of the full actual energy.

1.3 Organization

Section 2 defines the packet and its correct norm. The exact factor allocation is proved in section 3. The support criterion and explicit box certificate are in section 4. We transfer them to the sampled prime–squarefree packet in section 5. The source-energy obstruction and its sufficient repair are treated in section 6. Endpoint costs and the remaining gates are audited in section 7.

2 The declared retained-label packet

2.1 Rows, scales, and the static mask

Fix $h_0 \neq 0$, put $H = \operatorname{rad}|h_0|$, and take one source-compatible opened-d block. The source and opposite rows are

$$m = \ell_m d_m, \qquad m_\gamma = \ell_\gamma d_\gamma, \qquad \ell_m, \ell_\gamma \in [L, 2L], \quad d_m, d_\gamma \in [D, 2D],$$

where the ℓ's are prime, the d's are squarefree and coprime to H. Endpoint atoms may be assigned to either adjacent dyadic cell; every compact box selected below lies strictly inside these intervals. The inherited size separation makes each integer row determine its pair (ℓ, d) uniquely [2, 3]. Write

$$Q = LD, \qquad QJ \asymp X, \qquad J = K/D, \qquad c_X = \frac{LDJ}{X}, \quad c'_X = \frac{DJ}{K} = 1. \tag{1}$$

The literal static mask is

$$K_{m\gamma} = \mathbf{1}_{\ell_m \neq \ell_\gamma} \mathbf{1}_{|m - m_\gamma| > \Delta} \mathbf{1}_{(d_m, d_\gamma) \leq G}, \qquad \Delta = QX^{-\kappa_0}, \quad G = X^{\kappa_0}. \tag{2}$$

It remains literal throughout this paper.

Let $H_0 = O_{h_0}(1)$ denote the fixed orbit modulus and let $q = O_{h_0}(1)$ be a common multiple of every fixed row and orbit residue modulus. Fix one locally admissible source/opposite row-residue group modulo q and one orbit class $a \pmod{H_0}$. Direct splitting makes the periodic multiplier a constant ξ_a on this group and class. Cells with $\xi_a = 0$ are discarded. No Fourier normalization is attached to a direct cell.

The opposite-row universe is complete within its fixed dyadic and residue supports. Writing V for the inherited hard divisor cutoff, we assume the nondegenerate interior conditions

$$L > 4D, \qquad 2D \leq V, \qquad L, D \longrightarrow \infty, \tag{3}$$

and no independent opposite-row pruning. Together with the fixed positive-density unary and admissible residue support declared below, these conditions put the packet in the TPC–50 nondegenerate interior regime, where the coefficient-free opposite factor is a complete prime–squarefree cell and the literal mask survives [5].

2.2 The representative four-factor profile

The source coefficient is retained without alteration:

$$a_m := \gamma_m^{(1)} = \mu(d_m)(\log \ell_m)\omega_D(d_m)\psi_L(\ell_m/L)\zeta_m^{(1)}, \qquad |\zeta_m^{(1)}| \ll 1. \tag{4}$$

In particular, a_m is independent of the pre-terminal cofactor label r. The coefficient-free opposite row still has its fixed unary shape. We write it exactly as

$$v_0(\gamma) = \eta_\gamma R\left(\log \frac{\ell_\gamma}{L}, \log \frac{d_\gamma}{D}\right), \qquad |\eta_\gamma| = 1, \qquad R \in C^\infty([0, \log 2]^2), \tag{5}$$

on its nonzero direct support. Here R is the product of the fixed dyadic shapes and η_γ contains the Möbius sign and any remaining unit-modulus direct-cell phase. The exponent $e = 0$ removes the opposite prime logarithm; it does not replace R by one.

We now name the smooth product that was abbreviated by $\mathcal{W}$ in later interfaces. For fixed $W_1, W_2, \Psi_1, \Psi_2 \in C_c^\infty((0, \infty))$, put

$$\mathcal{W}_{m,\gamma}(j/J) := W_1(mj/X)\Psi_1(d_m j/K)W_2(m_\gamma j/X)\Psi_2(d_\gamma j/K). \tag{6}$$

This is the representative product obtained by placing one source-row factor and one opposite-row factor into the actual opened-row provenance [2]. Each has the form $W(mj/X)\Psi(dj/K)$. The same argument works verbatim for any fixed finite product; four factors keep the allocation visible.

Introduce normalized variables

$$\beta_m = (x_m, y_m, c_X, c'_X), \quad x_m = \log(\ell_m/L), \quad y_m = \log(d_m/D), \qquad q_\gamma = (x_\gamma, y_\gamma), \quad t_j = j/J. \tag{7}$$

Then the four arguments in (6) are

$$\begin{aligned} z_1(\beta, t) &= c_X \mathrm{e}^{x_m + y_m} t, & z_2(\beta, t) &= c'_X \mathrm{e}^{y_m} t, \\ z_3(\beta, q, t) &= c_X \mathrm{e}^{x_\gamma + y_\gamma} t, & z_4(\beta, q, t) &= c'_X \mathrm{e}^{y_\gamma} t. \end{aligned} \tag{8}$$

The normalized scale parameters range over a fixed compact set after a fixed dyadic subdivision.

2.3 The pre-terminal lift

Let $\mathcal{I}_X$ contain the retained pairs (m, r), let $\mathcal{O}_X$ be the complete opposite-row cell, and let

$$\mathcal{J}_{X,a} = \{ j : j/J \in I_0, j \equiv a \pmod{H_0} \},$$

where $I_0 \Subset (0, \infty)$ is one fixed interval containing the possible orbit support. Define

$$\mathcal{H}_X^{\mathrm{lift}} := \bigoplus_{(m,r) \in \mathcal{I}_X} \ell^2(\mathcal{J}_{X,a}; \ell^2(\mathcal{O}_X)). \tag{9}$$

For each outer summand, the map $e_\gamma \mapsto e_{\gamma, d_m r}$ is an isometry onto the physical coefficient subspace. Thus suppressing the deterministic residual coordinate here does not change the atomic normalization. The declared coefficient vector is

$$G_X^L(m, r, j) = \sum_{\gamma \in \mathcal{O}_X} a_m K_{m\gamma} \eta_\gamma R(q_\gamma) \xi_a \mathcal{W}_{m,\gamma}(j/J) e_\gamma. \tag{10}$$

Consequently

$$\boxed{\| G_X^L \|^2_{\mathcal{H}_X^{\mathrm{lift}}} = \sum_{m,r,j,\gamma} |a_m|^2 K_{m\gamma} |R(q_\gamma)|^2 |\xi_a|^2 |\mathcal{W}_{m,\gamma}(j/J)|^2.} \tag{11}$$

Every sum in (11) is over the retained sets just defined.

The label r is repeated orthogonally only in (9). The terminal residual vector later multiplies by $\mu(r)\Lambda(pr)$, sums coherently over p, and groups $s = d_m r$ [4]. Distinct (m, r) can then collide. Thus (11) is not an equality for that downstream norm.

This boundary can be written as an exact operator. Put $\mathcal{R}_X^{\mathrm{res}} := \{ d_m r : (m, r) \in \mathcal{I}_X \}$, let Φ_0 be the fixed inherited smooth terminal cutoff, and put $\Lambda(n) = (\log n)\Phi_0(n/X)$, and define

$$\mathcal{T}_{h_0} : \mathcal{H}_X^{\mathrm{lift}} \longrightarrow \ell^2(\mathcal{O}_X \times \mathcal{R}_X^{\mathrm{res}} \times \mathcal{J}_{X,a})$$

on an arbitrary lifted array $g = (g_{m,r,j,\gamma})$ by

$$(\mathcal{T}_{h_0} g)_{\gamma,s,j} := \sum_{\substack{m,r,p \\ s = d_m r \\ pr - mj = h_0}} \mu(r) \Lambda(pr) g_{m,r,j,\gamma}. \tag{12}$$

Here p is prime and every variable in the sum is restricted to the inherited finite terminal, row, cofactor, and orbit supports. With those supports understood, the equation $pr - mj = h_0$ determines the admissible terminal atoms. The TPC–48 physical left residual output is $\mathcal{T}_{h_0} G_X^L$.

Its later identification with the original large-prime residual carries one inherited global sign. Its norm is

$$\|\mathcal{T}_{h_0} G_X^L\|^2 = \sum_{\gamma,s,j} \left| \sum_{\substack{m,r,p \\ s=d_m r \\ pr-mj=h_0}} \mu(r)\Lambda(pr)[G_X^L(m,r,j)]_\gamma \right|^2, \tag{13}$$

and therefore contains cross terms absent from (11).

Cauchy–Schwarz on each output fiber gives the upper bound

$$\|\mathcal{T}_{h_0} g\|^2 \le \left(\sup_{\gamma,s,j} \sum_{\substack{m,r,p \\ s=d_m r \\ pr-mj=h_0}} |\mu(r)\Lambda(pr)|^2 \right) \|g\|^2_{\mathcal{H}_X^{\mathrm{lift}}}. \tag{14}$$

No corresponding lower bound follows from multiplicity alone: every output fiber containing two input atoms has a nontrivial kernel as a linear functional on unrestricted arrays. This is a logical non-implication, not a claim that the distinguished arithmetic array G_X^L belongs to that kernel. Moreover, the residual output itself belongs to the low-sign-averaged Gram route; its cross-prime inner products have not been identified with the native affine physical Gram [4].

3 The exact factor dictionary

3.1 Real-axis Mellin allocation

For $U \in C_c^\infty((0,\infty))$, use the convention

$$\widetilde{U}(\tau) = \int_0^\infty U(z) z^{-i\tau} \frac{dz}{z}, \qquad U(z) = \frac{1}{2\pi} \int_{\mathbb{R}} \widetilde{U}(\tau) z^{i\tau}\, d\tau. \tag{15}$$

Every transform is Schwartz. For $\mathbf{U} = (W_1, \Psi_1, W_2, \Psi_2)$, define the specific complex provenance measure

$$d\lambda(\boldsymbol{\tau}) = (2\pi)^{-4} \widetilde{W}_1(\tau_1)\widetilde{\Psi}_1(\tau_2)\widetilde{W}_2(\tau_3)\widetilde{\Psi}_2(\tau_4)\, d\boldsymbol{\tau}. \tag{16}$$

Write $d\lambda = \varepsilon\, d\mu$, where $\mu = |\lambda|$ and $|\varepsilon| = 1$ almost everywhere. No factor in (16) is moved into a row coefficient.

Choose $\chi \in C_c^\infty((0,\infty))$ equal to one on I_0. For one coordinate $i = (a, m, r, j, \gamma)$, set

$$\mathfrak{c}_X(i) := a_m K_{m\gamma} R(q_\gamma) \xi_a \chi(t_j) \tag{17}$$

and

$$\Phi_{\boldsymbol{\tau}}(i) := \eta_\gamma \varepsilon(\boldsymbol{\tau}) \prod_{\nu=1}^4 z_\nu(i)^{i\tau_\nu}. \tag{18}$$

All $z_\nu(i)$ are positive. Therefore $|\Phi_{\boldsymbol{\tau}}(i)| = 1$, and Mellin inversion gives

$$\int_{\mathbb{R}^4} \mathfrak{c}_X(i) \Phi_{\boldsymbol{\tau}}(i)\, d\mu(\boldsymbol{\tau}) = a_m K_{m\gamma} \eta_\gamma R(q_\gamma) \xi_a \prod_{\nu=1}^4 U_\nu(z_\nu(i)). \tag{19}$$

The auxiliary χ disappears because it is one on every retained coordinate.

Literal factor	Exact location	Reason
$\gamma_m^{(1)}$, as in (4)	outer source coefficient a_m	it is the single physical row coefficient in the left residual route
coefficient-free opposite row	base weight $\omega_{\gamma,X} = 1$; fixed shape $R(q_\gamma)$ in B, F; unit phase η_γ in the layer phase	$e = 0$ removes the logarithm, not the fixed unary shape
$K_{m\gamma}$	literal mask $M_X(m, \gamma)$	it is binary, so $K^2 = K$; no mask-layer parameter is introduced
ξ_a	both carrier B and physical profile F	direct splitting has no Fourier normalization
four Mellin transforms and $(2\pi)^{-4}$	complex measure λ, then positive measure $\mu = \lvert\lambda\rvert$	all remaining row/orbit powers are unimodular
cofactor r	repeated orthogonal pre-terminal label	a_m has no r-dependence before the terminal map
global residual sign	external fixed scalar when comparing with the original large-prime residual	it is absent from the declared physical vector and vanishes from squared identities
$\mu(r)\Lambda(pr)$, prime sum, and $s = d_m r$	not present	these are downstream operations

Table 1: Literal factor allocation for the declared left packet.

3.2 The allocation table

The entry $\omega_{\gamma,X} = 1$ deserves emphasis. In the original symmetric TPC–25 packet, the second row may carry its own physical prime logarithm. In the later left residual route, that row has become the opposite output coordinate; TPC–50 identifies its direct unary factor as the $e = 0$ case. Its fixed smooth factor R remains in both profiles, while η_γ is a unit layer phase; only the base logarithmic weight is one. The right residual route interchanges the two sides and again has $e = 0$ on its opposite row [2, 5].

3.3 Coordinatewise theorem

Define the carrier and physical profiles on normalized coordinates by

$$B_{\beta,a}(q, t) := \xi_a \chi(t) R(q), \tag{20}$$

$$F_{\beta,a}(q, t) := \xi_a \chi(t) R(q) W_1(z_1(\beta, t)) \Psi_1(z_2(\beta, t)) W_2(z_3(\beta, q, t)) \Psi_2(z_4(\beta, q, t)). \tag{21}$$

Theorem 3.1 (Exact actual-profile dictionary). *Fix the complete nondegenerate direct packet of section 2, with no independent opposite-row pruning. For every retained coordinate $i = (a, m, r, j, \gamma)$, take*

$$M_X(m, \gamma) = K_{m\gamma}, \qquad \omega_{\gamma,X} = 1, \qquad a_m = \gamma_m^{(1)}. \tag{22}$$

Then

$$|\mathfrak{c}_X(i)|^2 = M_X(m, \gamma)|a_m|^2 \omega_{\gamma,X} |B_{\beta_m,a}(q_\gamma, t_j)|^2, \tag{23}$$

$$|a_m|^2 K_{m\gamma} |R(q_\gamma)|^2 |\xi_a|^2 |\mathcal{W}_{m,\gamma}(t_j)|^2 = M_X(m, \gamma)|a_m|^2 \omega_{\gamma,X} |F_{\beta_m,a}(q_\gamma, t_j)|^2. \tag{24}$$

Both identities are exact for every X. Direct residue cells are disjoint, a_m is independent of r, and summing (24) gives precisely the lifted atomic norm (11).

Proof. Equation (17), the binary identity $K_{m\gamma}^2 = K_{m\gamma}$, and (20) give (23). Substitution of (8) in (6) gives (21), hence (24). Coordinatewise Mellin reassembly is (19) with the exact normalization (16). Direct orbit residue classes do not overlap, while m, r, j, γ are orthogonal labels in (9). Summation therefore gives (11) without a cross term or a counting factor. $\square$

Corollary 3.2 (Discharge of the preceding dictionary hypothesis)**.** *The exact-profile identification hypothesis of TPC–52 holds on the declared packet, with* $e = 0$*, in the retained-label pre-terminal coefficient space. Its complete physical factor is* $\eta_\gamma R(q_\gamma)\xi_a$ *times the named four-factor orbit product; the assertion* $e = 0$ *refers only to the logarithmic base weight.*

The scope is essential. The theorem does not identify an inherited packet whose $\mathcal{W}$ contains undeclared extra factors, an independently pruned opposite row, or a post-terminal coordinate grouping. Such a packet requires its own literal allocation table.

4 A deterministic support certificate

The dictionary is exact even when the physical profile is zero. A lower conditioning bound requires a separate overlap statement. We now make that statement finite-dimensional and checkable.

4.1 Continuous carrier and profile energies

Let $Y = [0, \log 2]^2$, with interior $Y^\circ = (0, \log 2)^2$, and

$$d\nu(q) = \mathrm{e}^{x+y}\, dx\, dy, \qquad q = (x, y). \tag{25}$$

This measure has mass one on Y. For a compact source-parameter set $\mathfrak{B}$, equipped with its relative topology in the normalized parameter domain, define

$$\mathscr{B}(\beta) := \frac{1}{H_0} \int_Y \int_{I_0} |B_{\beta,a}(q, t)|^2\, dt\, d\nu(q), \tag{26}$$

$$\mathscr{P}(\beta) := \frac{1}{H_0} \int_Y \int_{I_0} |F_{\beta,a}(q, t)|^2\, dt\, d\nu(q). \tag{27}$$

We discard a zero carrier cell. Compactness then gives

$$0 < b_- \le \mathscr{B}(\beta) \le b_+ < \infty \quad (\beta \in \mathfrak{B}) \tag{28}$$

after restricting to a nonzero compact source cell.

For a smooth function U, write

$$S_U := \{z > 0 : U(z) \neq 0\}. \tag{29}$$

Every S_U is open.

Definition 4.1 (Strictly reachable point)**.** A point $(\beta_0, q_0, t_0) \in \mathfrak{B} \times Y^\circ \times \operatorname{int} I_0$ is strictly reachable when

$$B_{\beta_0,a}(q_0, t_0) \neq 0, \qquad z_1 \in S_{W_1}, \quad z_2 \in S_{\Psi_1}, \quad z_3 \in S_{W_2}, \quad z_4 \in S_{\Psi_2}, \tag{30}$$

with every argument evaluated at (β_0, q_0, t_0).

For the two opposite-row arguments, strict reachability implies the useful ratio condition

$$\frac{z_3}{z_4} = \frac{c_X}{c'_X}\mathrm{e}^{x_0} \in \left(\frac{c_X}{c'_X}, 2\frac{c_X}{c'_X}\right). \tag{31}$$

At the TPC–25 scaling $c'_X = 1$, this is a direct comparison between the nonzero sets of W_2 and Ψ_2.

Theorem 4.2 (Reachable-support equivalence)**.** *For a fixed* β *satisfying* (28)*, the following are equivalent:*

(i) $\mathscr{P}(\beta) > 0$;

(ii) *there is a strictly reachable point with source parameter* β*;*

(iii) *there are compact boxes* $\mathcal{Q} \Subset Y^\circ$*,* $I \Subset \operatorname{int} I_0$*, and constants* $r_0, w_{1,0}, \psi_{1,0}, w_{2,0}, \psi_{2,0} > 0$ *such that throughout* $\mathcal{Q} \times I$*,*

$$|B_{\beta,a}| \geq r_0, \quad |W_i(z_{2i-1})| \geq w_{i,0}, \quad |\Psi_i(z_{2i})| \geq \psi_{i,0} \quad (i = 1, 2). \tag{32}$$

Proof. The integrand in (27) is continuous and nonnegative, and the measure $d\nu(q)dt$ is positive on every nonempty interior box. A positive integral therefore has a point where every factor is nonzero, proving (i) implies (ii). Openness of all nonzero sets and continuity give a compact product neighborhood and positive minima, which is (iii). Integrating the positive lower bound in (iii) proves (i). □

Theorem 4.3 (Explicit compact-box overlap)**.** *Suppose one strict point exists with* $\beta_0 \in \operatorname{relint} \mathfrak{B}$*. Then there are* $\mathfrak{B}_0 \Subset \mathfrak{B}$*,* $\mathcal{Q} \Subset Y^\circ$*, and* $I \Subset \operatorname{int} I_0$ *on which* (32) *holds uniformly in* $\beta \in \mathfrak{B}_0$*. For the full continuous energies, with the source parameter restricted to* $\mathfrak{B}_0$*,*

$$\boxed{\rho_* := \inf_{\beta \in \mathfrak{B}_0} \frac{\mathscr{P}(\beta)}{\mathscr{B}(\beta)} \geq \frac{r_0^2 w_{1,0}^2 \psi_{1,0}^2 w_{2,0}^2 \psi_{2,0}^2}{H_0 b_+} \left(\int_{\mathcal{Q}} \mathrm{e}^{x+y} \, dx \, dy \right) |I| > 0.} \tag{33}$$

Proof. Continuity in β extends the strict inequalities to a compact source box. The numerator of the profile energy is at least the integral over $\mathcal{Q} \times I$ of the product of the squared lower bounds. The carrier denominator is at most b_+. This gives (33). □

Thus the active source set

$$\mathfrak{B}_{\mathrm{act}} := \{\beta : \mathscr{P}(\beta) > 0\} \tag{34}$$

is relatively open. Every compact subset of it is uniformly active. A sequence of active parameters whose reachable arguments graze a cutoff boundary can lose its lower bound at the boundary; compactness of the full closed scale range alone therefore gives no positive uniform bound.

4.2 A legal nonempty model cell

The certificate is not vacuous. Take four nonnegative smooth cutoffs supported in $(1/2, 2)$ and equal to one on $[3/4, 5/4]$, take the fixed opposite unary shape R to equal one near the row point below, and take $c_X = c'_X = 1$. Choose I_0 to contain the point below. With

$$x_m = y_m = x_\gamma = y_\gamma = \log(21/20), \qquad t_0 = (21/20)^{-2},$$

the two W-arguments equal one and the two Ψ-arguments equal $20/21$. This is a strict interior point. It proves that the allowed smooth cutoff class contains certified packets; it does not assert that an unspecified inherited cutoff has these supports.

5 Transfer to a sampled prime–squarefree subpacket

Fix the compact boxes supplied by theorem 4.3, and write $\mathcal{Q} = [x_-, x_+] \times [y_-, y_+]$. Let $\mathcal{O}_X(\mathcal{Q})$ consist of all opposite rows in the fixed admissible residue group for which

$$q_\gamma = (\log(\ell_\gamma/L), \log(d_\gamma/D)) \in \mathcal{Q}.$$

This is a complete row universe inside a fixed interior interval, not an arbitrarily pruned subset. The fixed-modulus prime number theorem and squarefree counting in progressions give

$$|\mathcal{O}_X(\mathcal{Q})| = (c_{\mathcal{Q},\mathrm{loc}} + o(1)) \frac{LD}{\log L}, \qquad c_{\mathcal{Q},\mathrm{loc}} > 0, \tag{35}$$

and the normalized empirical row measure converges to the restriction of $d\nu = \mathrm{e}^{x+y}dxdy$ to $\mathcal{Q}$, divided by its mass [1, 5, 6].

The geometric mass before the fixed local-density factor is

$$p_{\mathrm{row}} = \nu(\mathcal{Q}) = (\mathrm{e}^{x_+} - \mathrm{e}^{x_-})(\mathrm{e}^{y_+} - \mathrm{e}^{y_-}) > 0. \tag{36}$$

For the retained orbit class,

$$\frac{1}{J}\#\{j : j/J \in I, j \equiv a \pmod{H_0}\} = \frac{|I|}{H_0} + O(J^{-1}). \tag{37}$$

Thus the row–orbit box has fixed geometric proportion

$$p_{\mathrm{geom}} = p_{\mathrm{row}}\frac{|I|}{H_0} > 0 \tag{38}$$

up to one fixed positive residue density.

The full complete-universe degree estimate from TPC–50 is

$$\sup_m \#\{\gamma : K_{m\gamma} = 0\} \leq \epsilon_X|\mathcal{O}_X|, \qquad \epsilon_X \ll_{h_0} L^{-1+o(1)} + D^{-1+o(1)} + X^{-\kappa_0+o(1)}. \tag{39}$$

Since (35) is a fixed positive fraction of the ambient count, restriction to $\mathcal{Q}$ changes the right side only by a fixed constant. In particular, the relative deletion inside the selected row box is still $o(1)$.

For a selected source row m, define the sampled carrier and profile energies

$$\mathscr{B}_X^M(m) := \sum_{\gamma \in \mathcal{O}_X(\mathcal{Q})} K_{m\gamma} \sum_{\substack{j/J \in I \\ j \equiv a \pmod{H_0}}} |B_{\beta_m,a}(q_\gamma, t_j)|^2, \tag{40}$$

$$\mathscr{P}_X^M(m) := \sum_{\gamma \in \mathcal{O}_X(\mathcal{Q})} K_{m\gamma} \sum_{\substack{j/J \in I \\ j \equiv a \pmod{H_0}}} |F_{\beta_m,a}(q_\gamma, t_j)|^2. \tag{41}$$

Theorem 5.1 (Sampled literal overlap). *Let every retained source parameter lie in $\mathfrak{B}_0$ from theorem 4.3. Then, for all sufficiently large X, the denominator below is nonzero and, uniformly in the source row,*

$$\boxed{\frac{\mathscr{P}_X^M(m)}{\mathscr{B}_X^M(m)} \geq \alpha_* > 0, \qquad \alpha_* := w_{1,0}^2\psi_{1,0}^2 w_{2,0}^2\psi_{2,0}^2.} \tag{42}$$

The same assertion holds for the symmetric right packet.

Proof. On the selected box the exact identity

$$|F_{\beta_m,a}(q_\gamma, t_j)|^2 = |B_{\beta_m,a}(q_\gamma, t_j)|^2 \prod_{i=1}^{2} |W_i(z_{2i-1})|^2 |\Psi_i(z_{2i})|^2$$

and (32) give the pointwise inequality $|F|^2 \geq \alpha_*|B|^2$. Multiplication by the nonnegative literal mask and summation prove the displayed ratio whenever its denominator is nonzero. Equations (35) and (37) provide a positive proportion of row–orbit coordinates, whereas (39) deletes only $o(1)$ of the rows for each source. Since $|B| \geq r_0$ on the box, at least one surviving coordinate remains for every sufficiently large X, so the denominator is positive. □

Let $\mathcal{U}_X$ be any finite set of retained (m, r) pairs in the selected source box, and let the same a_m repeat on its cofactor fibers. Put

$$\mathscr{C}_X = \sum_{(m,r) \in \mathcal{U}_X} |a_m|^2 \mathscr{B}_X^M(m), \tag{43}$$

$$\mathscr{Q}_X = \sum_{(m,r) \in \mathcal{U}_X} |a_m|^2 \mathscr{P}_X^M(m). \tag{44}$$

Corollary 5.2 (Certified interior coefficient bridge)**.** *If $\mathscr{C}_X > 0$, then*

$$\mathscr{Q}_X \geq \alpha_* \mathscr{C}_X. \tag{45}$$

For the provenance-canonical four-factor Mellin synthesis, using the notation $\mathcal{E}^L_{1,B}$ and $\mathcal{S}^L_X$ of TPC–52, the actual direction on the selected lifted packet has

$$\boxed{\frac{\mathcal{E}^L_{1,B}(\mathbf{1})^2}{\|\mathcal{S}^L_X \mathbf{1}\|^2} \leq \frac{C_B}{\alpha_*} = O(1),} \tag{46}$$

where C_B is the finite-cell carrier constant of TPC–52 and depends only on the fixed cutoffs, boxes, residue data, and derivative order.

Proof. Multiply (42) by $|a_m|^2$ on each cofactor fiber and sum. The exact dictionary theorem identifies $\mathscr{Q}_X$ with the squared physical lifted norm. The polar Mellin measure has fixed finite weighted variation, and every layer has carrier norm independent of its real Mellin parameter. The finite-cell carrier estimate of TPC–52 therefore gives $\mathcal{E}^L_{1,B}(\mathbf{1})^2 \leq C_B \mathscr{C}_X$. Combine the two bounds. □

No source equidistribution and no uniform cofactor multiplicity is used *inside the selected packet*. This statement is different from showing that the selected packet captures a fixed fraction of a larger packet, which is the subject of the next section.

6 Geometric coverage versus actual energy coverage

Let $\mathcal{M}^{\mathrm{full}}_X$ be a complete fixed-modulus source dyadic cell, and suppose the compatible cell $\mathfrak{B}_0$ contains a fixed positive-volume source rectangle. Let $\mathcal{M}^{\mathrm{good}}_X$ be the rows whose normalized parameters lie in that rectangle. Fixed-modulus prime and squarefree counting then gives

$$|\mathcal{M}^{\mathrm{good}}_X| \asymp |\mathcal{M}^{\mathrm{full}}_X|, \tag{47}$$

but the actual coefficient contains $\zeta^{(1)}_m$, for which the inherited interface assumes only an upper bound.

For the repeated lifted packet, define the fiber carrier multiplier

$$\mathfrak{n}_X(m) := \sum_{r:(m,r)\in\mathcal{I}_X} \mathscr{B}^M_X(m) = N_r(m)\mathscr{B}^M_X(m), \tag{48}$$

where $N_r(m)$ is the retained cofactor multiplicity. The good-carrier fraction in the full lifted packet is

$$\sigma_X := \frac{\displaystyle\sum_{m\in\mathcal{M}^{\mathrm{good}}_X} |a_m|^2 \mathfrak{n}_X(m)}{\displaystyle\sum_{m\in\mathcal{M}^{\mathrm{full}}_X} |a_m|^2 \mathfrak{n}_X(m)} \tag{49}$$

when the denominator is positive.

Proposition 6.1 (Sharp coefficient-concentration obstruction)**.** *Assume the full packet has at least one source row outside $\mathcal{M}^{\mathrm{good}}_X$ on which the fixed source shapes and fiber multiplier are nonzero. In the coefficient class $|\zeta^{(1)}_m| \leq 1$, one may have*

$$|\mathcal{M}^{\mathrm{good}}_X| \asymp |\mathcal{M}^{\mathrm{full}}_X| \qquad \textit{but} \qquad \sigma_X = 0. \tag{50}$$

Thus no positive lower bound for σ_X follows from geometric density, row sampling, or profile overlap alone.

Proof. Set $\zeta_m^{(1)} = 0$ on every good row and on any unwanted bad row, and set it equal to one on one nonzero bad row. This respects the only stated coefficient hypothesis. The numerator of (49) is zero and the denominator is positive. $\square$

Even a unit-modulus $\zeta_m^{(1)}$ does not by itself control σ_X if $N_r(m)$ is allowed to concentrate arbitrarily on bad rows. This is a second, logically independent coverage obstruction.

There is a clean sufficient repair. Let the fixed source unary shapes be bounded below on the good box, and suppose

$$|\zeta_m^{(1)}| \geq \zeta_0 > 0 \quad (m \in \mathcal{M}_X^{\text{good}}), \qquad |\zeta_m^{(1)}| \leq C \quad (m \in \mathcal{M}_X^{\text{full}}). \tag{51}$$

Proposition 6.2 (Source-energy coverage)**.** *Under* (51)*, fixed-modulus admissibility, uniform lower bounds for the source shapes on the good box, and uniform upper bounds for them on the full box, there is a constant* $c_{\text{src}} > 0$ *such that*

$$\frac{\sum_{m \in \mathcal{M}_X^{\text{good}}} |a_m|^2}{\sum_{m \in \mathcal{M}_X^{\text{full}}} |a_m|^2} \geq c_{\text{src}} + o(1). \tag{52}$$

If, in addition, for some $A_X > 0$ *and fixed* $0 < n_- \leq n_+ < \infty$*,*

$$n_- A_X \leq \mathfrak{n}_X(m) \leq n_+ A_X \quad \textit{on every nonzero full source row,} \tag{53}$$

then

$$\sigma_X \geq \frac{n_-}{n_+} c_{\text{src}} + o(1) > 0. \tag{54}$$

Proof. On the good box, the lower bounds for the fixed source shapes and $\zeta_m^{(1)}$, together with the fixed-modulus prime number theorem and squarefree counting, give

$$\sum_{m \in \mathcal{M}_X^{\text{good}}} |a_m|^2 \gg LD \log L.$$

The inherited upper bounds and Chebyshev's estimate give $\sum_{m \in \mathcal{M}_X^{\text{full}}} |a_m|^2 \ll LD \log L$. This proves (52). Multiplying the good numerator by the lower bound in (53) and the full denominator by the upper bound gives (54). $\square$

Condition (53) is not proved by the present paper. It is a transparent sufficient condition for turning source-row coverage into coverage after cofactor repetition. Alternatively, a direct estimate of (49) may replace both (51) and (53).

7 Endpoint ledger, limitations, and next gates

At the inherited endpoint,

$$\begin{aligned} Q &= X^{267/400+o(1)}, \qquad J = X^{133/400+o(1)}, \quad \kappa_0 = \frac{1}{400}, \\ D &= X^{10049/52500+o(1)}, \quad V = X^{23/120+o(1)}. \end{aligned} \tag{55}$$

The hard divisor boundary has the strict margin

$$\frac{23}{120} - \frac{10049}{52500} = \frac{9}{35000} > 0. \tag{56}$$

Moreover $L = Q/D$ and

$$\left(\frac{267}{400} - \frac{10049}{52500}\right) - \frac{10049}{52500} = \frac{59783}{210000} > 0. \tag{57}$$

Hence $2D \leq V$ and $L > 4D$ for all sufficiently large X, each with a fixed power margin. Together with the certified unary and residue support, the selected packet is genuinely nondegenerate; no endpoint loss is hidden in these conditions [5].

The costs of the present extraction are:

(E1) a fixed number of direct row and orbit residue cells;

(E2) fixed compact source, opposite-row, and orbit boxes;

(E3) row sampling error $o(1)$ at fixed modulus;

(E4) orbit sampling error $O(J^{-1})$; and

(E5) literal-mask deletion

$$O\Big(L^{-1+o(1)} + D^{-1+o(1)} + X^{-1/400+o(1)}\Big).$$

Every Mellin transform belongs to a fixed Schwartz library and has finite weighted variation. Since the support certificate gives a fixed $\rho_* > 0$, only qualitative $o(1)$ sampling is required. Thus the local dictionary-and-overlap bridge has zero positive polynomial cost and spends none of the $1/400$ allowance.

This zero cost applies to the *selected* packet. It does not say that deleting the complement costs zero actual energy. Under arbitrary bounded $\zeta_m^{(1)}$ or arbitrary cofactor multiplicity, the complement may carry all of the lifted carrier energy by proposition 6.1. A full-packet statement must prove a positive σ_X, partition all nonzero energy into finitely many compatible cells, or bound the unselected complement by another method.

7.1 Exact claim ledger

Proved exactly (L0).
The exact four-factor Mellin allocation and lifted atomic norm. Also, the reachable-support equivalence, explicit compact-box lower bound, and source-concentration obstruction.

Proved on the literal selected packet (L1).
The exact TPC–52 profile dictionary with $e = 0$, fixed positive geometric row–orbit density, sampled masked overlap, and an $O(1)$ canonical condition number in the retained-label pre-terminal space.

Conditional subgate.
Positive coverage of the full lifted carrier follows from a nonvanishing source factor plus balanced carrier multiplicities, or from any direct proof that $\sigma_X \gg 1$.

Not proved (L2).
A coherent-prime lower square, a fixed-h_0 arithmetic lower bound, a parity-sensitive estimate, or a twin-prime theorem.

7.2 Dependency order after this paper

The remaining interfaces should not be permuted:

(O1) determine whether compatible cells capture a controlled fraction of the full actual carrier energy, including cofactor multiplicity;

(O2) prove, or sharply delimit, a coherent prime square at fixed (m, r) that is stable under the certified carrier projection;

(O3) control the many-to-one grouping $s = d(m)r$ and its cancellation;

(O4) localize a completed frozen shift family back to the single physical intercept h_0; and

(O5) compare the residual low-sign Gram with the native affine physical Gram without deleting its cross terms.

The present paper advances the first physical-interface line by replacing an abstract factor dictionary with an exact one and replacing an unnamed overlap assumption with a finite support certificate. It does not move any downstream operation through a norm and does not hint at a resolution of the parity problem.

References

[1] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI, 2004.

[2] Liang Wang. Zero before separation in a primitive Möbius tail: Provenance-weighted closure and the principal row-mode boundary. Preprint, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-25-zero-before-separation.

[3] Liang Wang. Soft projective reassembly of the physical row mask: Orthogonal GCD compression, optimal ferrers separation, and coherent product-slope spectra. Preprint, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-36-soft-projective-mask-reassembly.

[4] Liang Wang. Critical orbit tiling and an all-band large-sieve reduction for the actual-coefficient residual determinant family. Preprint, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-48-critical-orbit-tiling-large-sieve.

[5] Liang Wang. Fixed-complexity cell survival and sharp concentration obstructions for the TPC static mask. Preprint, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-50-fixed-complexity-cell-survival.

[6] Liang Wang. Canonical mellin quotient frames: Finite-sampling kernels, interior stability, and the boundary of signed TPC reassembly. Preprint, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-51-canonical-mellin-quotient-frames.

[7] Liang Wang. Diagonal mellin reassembly on actual orbit fibers: Scalar saturation, profile overlap, and an interior TPC bridge. Preprint, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-52-diagonal-mellin-orbit-fibers.